Operating Systems: Deadlock and Process Synchronization

1. What is a Deadlock?

A deadlock is a situation where processes are blocked waiting for each others resources.

2. Necessary Conditions for Deadlock - Coffman Conditions:

- Mutual Exclusion: At least one resource must be non-shareable.
- Hold and Wait: A process holds resources while waiting for additional ones.
- No Preemption: Resources cannot be preempted forcibly.
- Circular Wait: A closed loop of processes waiting for each other.

3. Deadlock Handling Strategies:

- Prevention: Invalidate one of Coffman conditions.
- Avoidance: Use Bankers Algorithm to ensure safe state.
- Detection and Recovery: Resource Allocation Graph and termination.

4. Critical Section and Mutex:

Critical section is shared memory access. Mutex ensures mutual exclusion.